

sol2_implementer_agent_sdk

Take-home **configuration port**. It does not run the eight-step loop.

This is the role graph. The live harness is `sol2_implementer_deep_agents`.

It prints the role table and builds `ClaudeAgentOptions`. There is no `implementer.py`, no `gates.py`, and no `rubric.py`. The driver lives in the Deep Agents twin. There is a `tests/` now, and it is what keeps the fence honest.

What it proves

The role table survived the runtime swap.

role	writes	scope	
orchestrator	no	nothing	
planner	yes	steps.jsonl	
test_implementer	yes	tests/**	
code_implementer	yes	app/**, src/**	denied tests/**
judge	no	nothing	

Do not copy this folder into `labs/lab2_implementer/harness.py`. The Saturday stub wants `red_gate / score_attempt / run_loop`. This file exports `cast / build / backend`.

Learning objectives

- Read `roleplan.LOOPS["implementer"]`
- Wire `tools=[...]` plus `PreToolUse`
- Print `--table-only` with no SDK and no key
- Pair this port with `sol2_implementer_deep_agents` when you want the eight steps to run

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Starting architecture

```
python3 harness.py --table-only
└─ roleplan.plan(contract, "implementer")
    five RolePlans, judge can_write False

python3 harness.py --repo CRM
└─ ClaudeAgentOptions
    agents for planner, test_implementer, code_implementer, judge
    permission_mode=dontAsk
    one PreToolUse scope_hook for the whole cast
```

No call to `implementer.run`.

Scope. Two places, both required

`tools=[...]` decides whether a role can write at all.

`scope_hook` decides which paths. Deny envelope:

```
hookSpecificOutput.hookEventName = PreToolUse
hookSpecificOutput.permissionDecision = deny
```

A typo fails **open**. The field is `maxTurns`, camelCase: `max_turns` raises `TypeError` on the real SDK, and a test fake taking `**kwargs` hides it.

Commands

```
cd solutions/sol2_implementer_agent_sdk
python3 harness.py --table-only
python3 harness.py --repo ../../work/northwind-field-crm
task test    # pytest. task table is --table-only
```

`--table-only` with a missing repo still prints the table and exits 0.

Expected table

Judge must print `no`. Code implementer allow `app/**`, `src/**`, `deny tests/**`.

If judge prints `yes`, the port is wrong. Stop.

Troubleshooting

SYMPTOM	CAUSE	FIX
Thought this ran T001	config port	use Deep Agents harness.py
Judge writes yes	tools leaked Edit	strip judge tools
Fail-open writes	deny shape typo	full hookSpecificOutput
Copied into lab2 stub	wrong artifact	Saturday wants three functions

Validation

- [] table: five roles, judge no
- [] no `implementer.py` in this folder (on purpose)
- [] `task test` needs no key, no SDK, and no clone
- [] you can name the folder that actually runs the loop

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Recap

Config port, not a filled loop. Same table, different enforcement knob.

The eight steps live in `sol2_implementer_deep_agents`. This folder proves the table survived.

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Prerequisites

```
cd solutions/sol2_implementer_agent_sdk  
python3 -c "import roleplan, roles, write_scope; print('ok')"
```

No API key. No CRM clone. `claude-agent-sdk` is only needed if you call `options_for` without stubbing.

Saturday Lab 2 still fills `labs/lab2_implementer/harness.py`. This folder is take-home.

Files in this folder

```
SPEC.md Taskfile.yml  
adapter.py contract.py harness.py  
roleplan.py roles.py write_scope.py
```

Missing on purpose: `implementer.py`, `gates.py`, `rubric.py`, `doers.py`, `tests/`.

Same table, three enforcements

RUNTIME

Saturday Claude

Agent SDK (this folder)

Deep Agents

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HOW JUDGE CANNOT WRITE

YAML tools omit Write

tools omit Edit/Write, plus PreToolUse

subagent tools = [read_file]

Final checklist

- ☐ `--table-only` prints judge `no`
- ☐ `code_implementer` deny includes `tests/**`
- ☐ you did not copy this `harness.py` over the Saturday stub
- ☒ you know `sol2_implementer_deep_agents` is the live loop

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